

Module 5: Persistent Trapping and Slow Recovery at Cryogenic Temperature

Purpose of this module

This module converts the cryogenic kinetics into a concrete device-level consequence. The central claim is that radiation history can persist longer at low temperature because trap emission and annealing become too slow to erase that history on operational timescales.

The main model

The project models trapped-charge recovery as

$$Q_{\text{trap}}(t, T) = Q_0 \exp\left(-\frac{t}{\tau_{\text{rec}}(T)}\right), \quad (1)$$

where Q_0 is the initial trapped charge after irradiation and $\tau_{\text{rec}}(T)$ is the recovery time constant.

Why exponential recovery is reasonable

Exponential recovery is the standard solution of a first-order relaxation law. Suppose trapped charge leaves the trapped population at a rate proportional to how much trapped charge remains:

$$\frac{dQ_{\text{trap}}}{dt} = -\frac{1}{\tau_{\text{rec}}} Q_{\text{trap}}. \quad (2)$$

Separating variables and integrating gives

$$\int \frac{dQ_{\text{trap}}}{Q_{\text{trap}}} = -\int \frac{dt}{\tau_{\text{rec}}}, \quad (3)$$

$$\ln Q_{\text{trap}} = -\frac{t}{\tau_{\text{rec}}} + C, \quad (4)$$

so

$$Q_{\text{trap}}(t, T) = Q_0 e^{-t/\tau_{\text{rec}}(T)}. \quad (5)$$

Analogy

Think of popcorn kernels escaping from a bowl through a small hole. If at every moment the escape rate is proportional to how many kernels remain, you get exponential emptying. Here the trapped charge population is the bowl, and the temperature-dependent recovery time controls how fast the hole effectively works.

Connecting recovery time to cryogenic kinetics

The recovery time is modeled with an Arrhenius law such as

$$\tau_{\text{rec}}(T) = \tau_{\text{ref}} \exp\left(\frac{E_r}{k_B T}\right). \quad (6)$$

This form is the reciprocal of an activated rate. As temperature decreases, the time constant grows rapidly.

Physical interpretations of τ_{rec}

Depending on what mechanism dominates, the recovery time may represent:

- thermal emission from charge traps,
- structural recovery through annealing,
- an effective combination of both.

The exact microscopic interpretation can vary, but the project-level message stays the same: lower temperature means longer persistence.

Time to 50% recovery

A useful metric for plots and engineering discussion is the half-recovery time. Set

$$Q_{\text{trap}}(t_{50}, T) = \frac{Q_0}{2}. \quad (7)$$

Then

$$\frac{1}{2} = e^{-t_{50}/\tau_{\text{rec}}}, \quad (8)$$

$$\ln 2 = \frac{t_{50}}{\tau_{\text{rec}}}, \quad (9)$$

so

$$t_{50}(T) = \tau_{\text{rec}}(T) \ln 2. \quad (10)$$

This is a very convenient summary because it directly shows how quickly radiation history is forgotten at each temperature.

Why this matters to devices

Persistent trapped charge can affect device behavior in several ways:

- baseline offsets can remain longer,
- threshold-like shifts can last longer,
- charge collection recovery can be delayed,
- time-dependent noise or drift may carry memory of prior radiation exposure.

The project does not need to model every transistor detail to make the core point. Even a simple trapped-charge persistence law already shows why low-temperature operation can preserve radiation signatures.

Connecting the whole cryogenic story

Modules 4 and 5 together tell a clean progression:

1. cooling suppresses migration,
2. cooling suppresses trap emission,
3. cooling suppresses annealing,
4. therefore trapped charge and damage signatures can persist,
5. therefore devices can remember radiation history for longer.

What this means operationally

At room temperature, some damage signatures can partially relax during ordinary operation. At cryogenic temperature, that same recovery can become so slow that the device effectively stores a memory of irradiation over the time window relevant to the system.

What to remember from this module

- Exponential recovery comes from a first-order relaxation equation.
- The recovery time constant grows rapidly as temperature decreases.
- Half-recovery time is a simple engineering metric: $t_{50} = \tau_{\text{rec}} \ln 2$.
- Cryogenic devices may retain radiation history much longer than room-temperature intuition suggests.